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$u = 5 \times 10^5$ km/h

$a = 1000\ m\,s^{-2}$

$v = 2u$

$a = \frac{v-u}{t}$

$t = \frac{v-u}{a}$

$5 \times 10^5 \times 1000$ m/hr

5×10^8 m/hr

$\frac{5 \times 10^8}{3600} = 138888\ m\,s^{-1}$

$u = 138888\ m\,s^{-1}$

$v = 2(138888)\ m\,s^{-1}$

$= 277776\ m\,s^{-1}$

1) $u = 138888$

$v = 277776$

$a = 1000\ m\,s^{-2}$

2) $t = ?$

3) $t = \frac{v-u}{a}$

4) $t = \frac{277776 - 138888}{1000}$

$t = 139$ s

① Derive $v = u + at$

By Definition of $a = \frac{v-u}{t}$

$at = v - u$

$v - u = at$

$+u \quad +u$

$v = u + at$

② Derive $s = ut + \frac{1}{2}at^2$

average velocity $= \frac{u+v}{2}$ Velocity $= \frac{\text{Displacement}}{\text{Time}}$

$\frac{u+v}{2} = \frac{s}{t}$

$\frac{s}{t} = \frac{u+v}{2}$

$s = \frac{(u+v)t}{2}$ → $v = u + at$

$s = \frac{(u+u+at)t}{2}$

$s = \frac{2ut + at^2}{2}$ ÷2

$s = ut + \frac{1}{2}at^2$

$s = \frac{2ut}{2} + \frac{at^2}{2}$

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$v = u + at$

$s = ut + \frac{1}{2}at^2$

$v^2 = u^2 + 2as$

HW Activity Book Pg 12 Q7–9.

① $u = 0$

$v = 14\ ms^{-1}$

$t = 7\ s$

② $a = ?$

$s = ?$

③ $v = u + at$

$a = \frac{v - u}{t}$

④ $a = \frac{14 - 0}{7}$

⑤ $a = 2\ ms^{-2}$

① $u = 0$

$v = 14\ ms^{-1}$

$a = 2\ ms^{-2}$

$t = 7\ s$

② $s = ?$

③ $s = ut + \frac{1}{2}at^2$

④ $s = (0)(7) + \frac{1}{2}(2)(7)^2$

⑤ $s = 49\ m$